

COMPLETE UPPER LIMB + SPINE REVIEWER (MIDTERM LEVEL)

I. CORE BIOMECHANICS PRINCIPLES (MUST KNOW)

1. Osteokinematics

Visible bone movement:

- Flexion / Extension
 - Abduction / Adduction
 - Rotation
-

2. Arthrokinematics

Joint surface movement:

- Roll
 - Glide
 - Spin
-

3. Concave–Convex Rule

- Convex on concave → glide opposite direction
 - Concave on convex → glide same direction
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4. Open vs Closed Pack

- Open-pack = loose joint
 - Close-pack = stable joint
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II. NERVE ROOT MAP (UPPER LIMB)

- C5 → shoulder abduction, elbow flexion
 - C6 → wrist extension
 - C7 → elbow extension
 - C8 → finger flexion
 - T1 → finger abduction/adduction
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III. SPINE (COMPLETE)

1. Regions

- Cervical (7)
 - Thoracic (12)
 - Lumbar (5)
 - Sacrum
 - Coccyx
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2. Vertebra Structure

- Body
 - Pedicle
 - Lamina
 - Spinous process
 - Transverse process
 - Vertebral foramen
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3. Special Vertebrae

- C1 (Atlas) → no body
 - C2 (Axis) → dens
 - C7 → vertebra prominens
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4. Curvatures

- Cervical lordosis
 - Thoracic kyphosis
 - Lumbar lordosis
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5. Intervertebral Disc

- Annulus fibrosus
- Nucleus pulposus

Function:

- Shock absorber
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6. Ligaments

- ALL → prevents hyperextension

- PLL → prevents flexion
 - Ligamentum flavum → elastic recoil
 - Supraspinous
 - Interspinous
 - Intertransverse
 - Ligamentum nuchae
-

7. Spinal Cord

Ends: L1–L2

Structures:

- Conus medullaris
 - Cauda equina
-

8. Meninges & Spaces

- Epidural → fat + veins
 - Subdural → potential space
 - Subarachnoid → CSF
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9. Spine Movements

- Flexion
 - Extension
 - Rotation
 - Lateral flexion
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IV. BACK MUSCLES (IMPORTANT OINA)

1. Trapezius

- O: skull, C7–T12
 - I: clavicle, scapula
 - A: elevate/retract scapula
 - N: CN XI
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2. Latissimus dorsi

- O: T7–L5
 - I: humerus
 - A: extension/adduction
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- N: thoracodorsal
-

3. Erector spinae

- Iliocostalis
 - Longissimus
 - Spinalis
 - A: extension
 - N: posterior rami
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4. Multifidus

- Stabilizer of spine
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V. ELBOW (COMPLETE)

1. Joints

- Humeroulnar → hinge
 - Humeroradial
 - Proximal radioulnar → pivot
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2. Osteokinematics

- Flexion/Extension
 - Pronation/Supination
-

3. Arthrokinematics

- Ulna on humerus:
 - Flexion → anterior glide
 - Extension → posterior glide
 - Radius spins in annular ligament
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4. Carrying Angle

- 5–15° valgus
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5. Close Pack

- Full extension + supination

6. Open Pack

- ~70° flexion

7. Key Muscles (OINA)

Biceps brachii

- Flexion + supination
- Musculocutaneous

Brachialis

- Main flexor

Triceps

- Extension
- Radial nerve

Pronators

- Pronator teres
- Pronator quadratus

Supinators

- Supinator
- Biceps

8. Nerves

- Musculocutaneous → flexors
- Radial → extensors
- Ulnar → medial hand
- Median → anterior forearm

VI. WRIST (COMPLETE)

1. Carpal Bones

Proximal:

- Scaphoid
- Lunate
- Triquetrum
- Pisiform

Distal:

- Trapezium
- Trapezoid
- Capitate (**os magnum**)
- Hamate

2. Joints

- Radiocarpal
- Midcarpal

3. Movements

- Flexion
- Extension
- Radial deviation
- Ulnar deviation

4. Arthrokinematics

- Carpals convex on radius
- Flexion → dorsal glide
- Extension → volar glide

5. Carpal Tunnel

Contents:

- 9 flexor tendons
- Median nerve

NOT included:

- FCR
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VII. FLEXOR TENDON SYSTEM (HAND)

1. Flexor Tendons

- FDS → PIP
 - FDP → DIP
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2. Pulley System

Annular pulleys:

- A1 → MCP
 - A2 → proximal phalanx (MOST IMPORTANT)
 - A3 → PIP
 - A4 → middle phalanx (VERY IMPORTANT)
 - A5 → DIP
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Cruciate pulleys:

- allow flexibility
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3. Function

- Prevent bowstringing
 - Maintain mechanical efficiency
-

4. Clinical

- A2/A4 rupture → bowstringing
 - A1 → trigger finger
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VIII. FLEXOR TENDON ZONES

Zone I

- DIP only (FDP injury)

Zone II

- FDS + FDP
- "No man's land"

Zone III

- Palm

Zone IV

- Carpal tunnel

Zone V

- Forearm
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IX. EXTENSOR SYSTEM

Extensor expansion (hood)

- central slip → middle phalanx
 - lateral bands → distal phalanx
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Clinical

- Mallet finger → DIP
 - Boutonnière → PIP
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X. HAND (COMPLETE)

1. Thenar (median)

- Opposition
 - Precision grip
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2. Hypothenar (ulnar)

- Power grip
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3. Lumbricals

- MCP flexion
 - IP extension
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4. Interossei

- Dorsal → ABduct (DAB)
 - Palmar → ADduct (PAD)
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5. Nerves

- Median → lateral hand
 - Ulnar → intrinsic muscles
 - Radial → extension only
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6. Functional Summary

- Precision = median
 - Power = ulnar
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